

$$u_{12} = \frac{d\Phi}{dt} \quad (\Phi = \int_A \vec{B} \cdot d\vec{A})$$

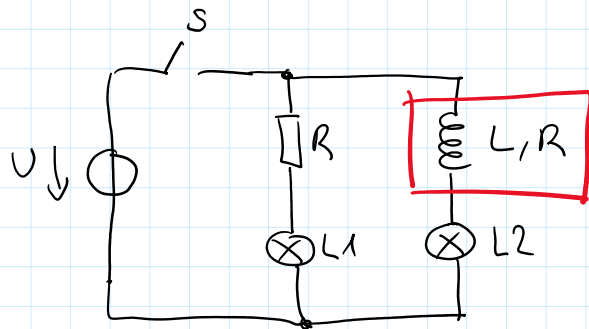
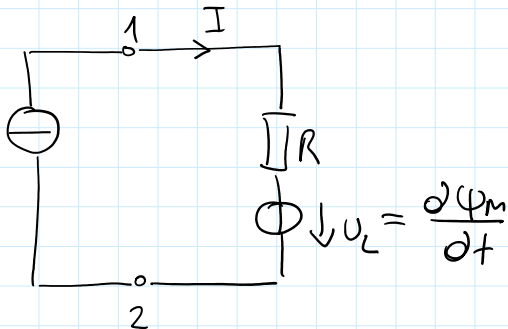
Messung N -Windungen

$$u_{12} = N \cdot \frac{d\Phi}{dt}$$

$$\Psi_m = \sum_{k=1}^N \Phi_k$$

$$\oint \vec{E} \cdot d\vec{s} = - \frac{d}{dt} \int_A \vec{B} \cdot d\vec{A}$$

(2. Maxwell'sche Gleichung)



DC $\frac{d\psi_m}{dt} = 0 \quad u_{12} = R \cdot I$

AC $u_{12} = R \cdot i + \frac{d\psi_m}{dt}$

$\frac{di}{dt} > 0 \quad u_L = \frac{d\psi_m}{dt} > 0 \rightarrow \text{Verbraucher}$

$\frac{di}{dt} < 0 \quad u_L < 0 \rightarrow \text{Erzeuger}$

$L = \frac{\psi_m}{I} \quad [L] = 1 \text{ Henry} = 1 \text{ H}$

$\psi = N \cdot \Phi \Rightarrow L = N \cdot \frac{\Phi}{I}$

$u_L = L \cdot \frac{di}{dt} \quad (u = \frac{d\psi_m}{dt})$

Φ

$$U_L = L \frac{dI}{dt} \quad (U = \frac{d\Phi}{dt})$$

IDEALISIERTER TRANSFORMATOR

- verlustlos
- magnetischer Widerstand = 0
- $U_1 \dots$ sinusförmige Spannung

$$\Phi = \Phi(t) = \Phi_{\max} \sin(\omega t)$$

$$\hat{U}_1 = N_1 \cdot \omega \cdot \Phi_{\max}$$

$$U_1 = \frac{2\pi f N_1 \Phi_{\max}}{\sqrt{2}} = 4,44 f N_1 \Phi_{\max}$$

$$U_2 = 4,44 f N_2 \Phi_{\max}$$

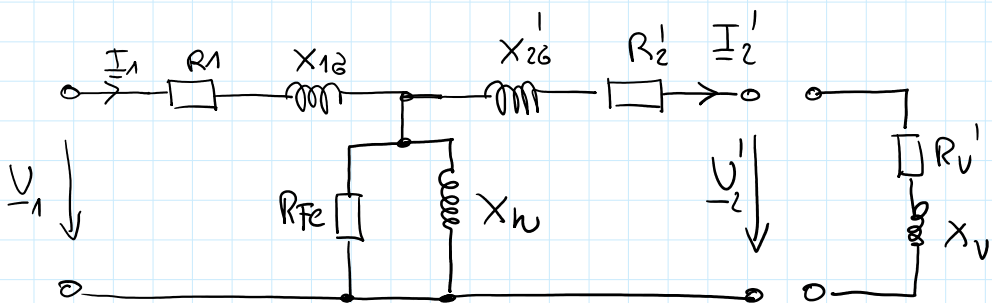
$$\frac{U_1}{U_2} = \frac{N_1}{N_2} = \underline{\underline{U}}$$

$$\frac{I_1}{I_2} = \frac{U_2}{U_1} = \frac{N_2}{N_1} = \frac{1}{\underline{\underline{U}}}$$

$$\text{da } S_1 = S_2 \quad U_1 \cdot I_1 = U_2 \cdot I_2$$

UNTERSCHIED REALER $\leftrightarrow$ IDEALISIERTER TRAF

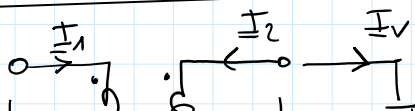
- Streng magnetischer Fluss
- Stromwiderstand und Eisenverluste
- Magnetisierbarkeit
- Ersatzschaltbild



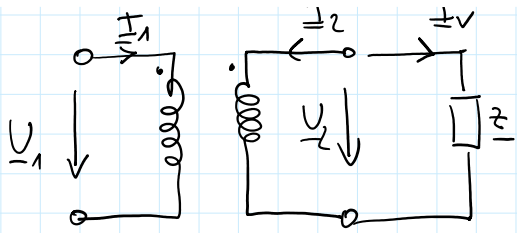
$$R_2' = \underline{\underline{U}}^2 R_2 \quad X_{22}' = \underline{\underline{U}}^2 X_{22}$$

$$U_2' = \underline{\underline{U}} U_2 \quad I_2' = \frac{I_2}{\underline{\underline{U}}}$$

Neuerkennungsfunktion

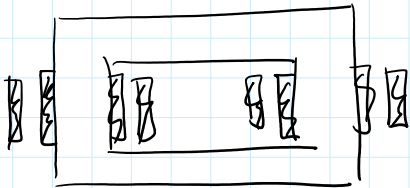


$$\underline{\underline{z}} = \frac{U_1}{I_1} = \frac{\underline{\underline{U}}^2 U_2}{-I_2} = \underline{\underline{U}}^2 \frac{U_2}{I_2} = \underline{\underline{U}}^2 \underline{\underline{z}}$$

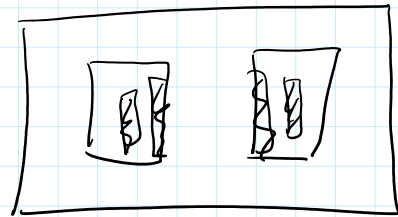


$$\underline{z_e} = \frac{\underline{U_1}}{\underline{I_1}} = \frac{U \underline{U_2}}{-\underline{I_2}} = \underline{U}^2 \frac{\underline{U_2}}{\underline{I_2}} = \underline{U}^2 \underline{z}$$

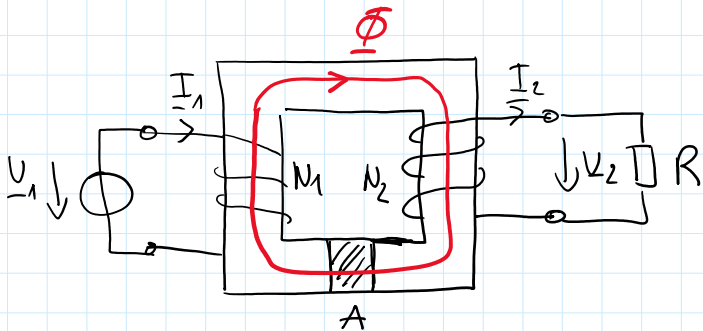
KERNTYP



FLAUTENTYP



BSP



$$N_1 = 575$$

$$R = 5 \Omega$$

$$N_2 = 60$$

$$A = 1600 \text{ mm}^2$$

ges. $\hat{B}, I_1, I_2$

$$U_1 = 230 \text{ V}$$

$$f = 50 \text{ Hz}$$

$$U_1 = 4,44 \cdot f \cdot N_1 \cdot \hat{\Phi} \quad \hat{\Phi} = A \cdot \hat{B}$$

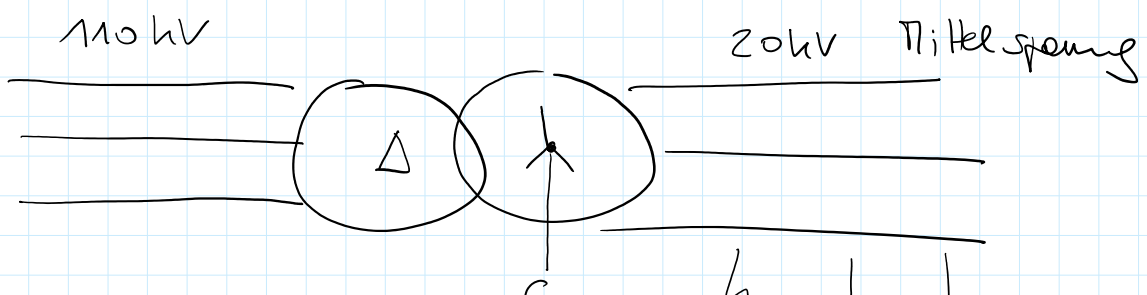
$$U_1 = 4,44 \cdot f \cdot N_1 \cdot A \cdot \hat{B}$$

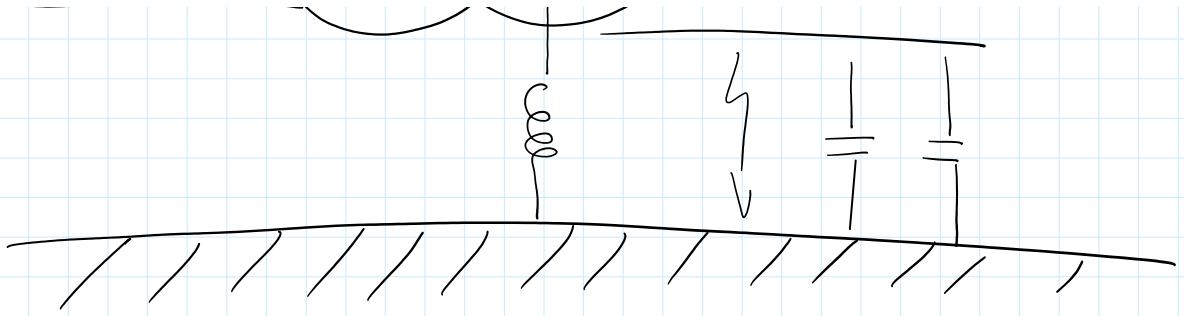
$$\hat{B} = \frac{U_1}{4,44 f N_1 A} = \frac{230 \text{ V}}{4,44 \cdot 50 \text{ Hz} \cdot 575 \cdot 1600 \cdot 10^{-6} \text{ m}^2} = 1,13 \text{ T}$$

$$\underline{U_2} = U_1 \cdot \frac{N_2}{N_1} = 230 \text{ V} \cdot \frac{60}{575} = 24 \text{ V}$$

$$I_2 = \frac{U_2}{R} = \frac{24 \text{ V}}{5 \Omega} = 4,8 \text{ A}$$

$$I_1 = I_2 \cdot \frac{N_2}{N_1} = 4,8 \text{ A} \cdot \frac{60}{575} = 0,5 \text{ A}$$





▷ NOSPE Niedrige Skulpturleistung

▷ STARRE

▷ isoliert

▷ gelichtet

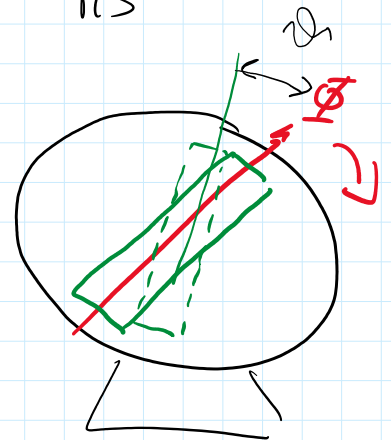
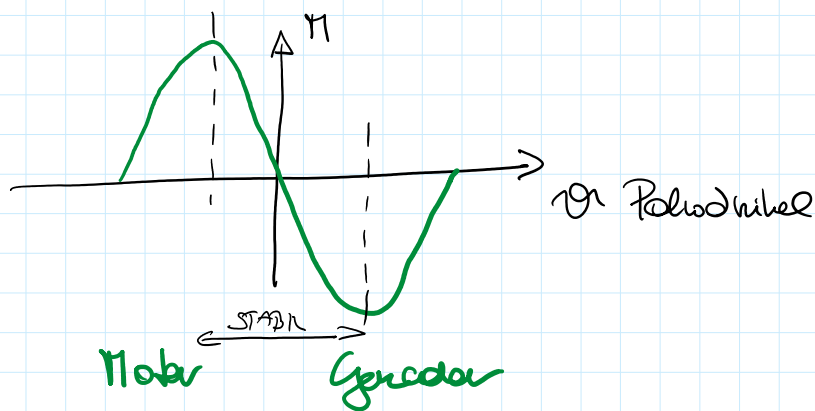
NS

NS, π S

π S

π S

SYNCHRONMASCHINE



$$n = \frac{p}{P} \text{ [1/s]}$$

P... Polpaarzahl

Gostendine

$p=1$

Wasserkraft

$p>$

Windkraft

$p>>$